LESSON PLAN

GUN SYSTEM (35)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Propulsion system used in torpedo, power supply and control systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Propulsion system used in torpedo, power supply and control systems
5. Specific Objectives	35.1.1.1 Trainee will be able to explain role of underwater weapons

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propulsion diagrams and control circuit schematics; Safety: No live electrical components; Instructor supervision: Monitor diagram interpretation accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a torpedo?
(b) Name two types of underwater weapons.
(c) How does a torpedo differ from a surface missile?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of torpedo propulsion systems • Chemical fuel cell operation principles • Electric motor drive mechanisms • Power supply battery configurations • Control system actuation methods	L & Ds	PPT, IB	15 min
(b)	• Thermal battery activation sequences • Hydraulic control line functions • Gyroscopic stabilization systems • Depth charge deployment mechanisms • Propeller pitch adjustment controls	L & Ds	PPT, PDF	20 min
(c)	• Integration of power and control units • Fail-safe mechanisms in propulsion • Signal transmission in control circuits • Maintenance requirements for power cells • Troubleshooting common control faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the thermal battery in torpedo propulsion.
(b) Describe how the control system directs the torpedo's path.
(c) Identify the primary power source for modern electric torpedoes.

12. Home Assignment and Follow Up	Prepare a brief note on the guidance system suitable for underwater weapons to be discussed in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Guidance system suitable for under water weapons	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min